Kinematic structure

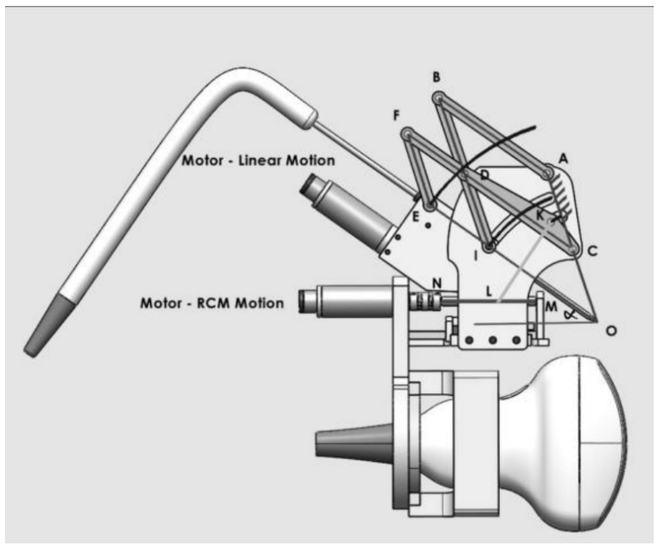


Fig. 2. Kinematic linkages of the proposed design.

The double parallelogram guides the needle axis through the remote center O. Moving L along MN changes the inclination; insertion along the needle axis is driven separately.

Workspace of the needle

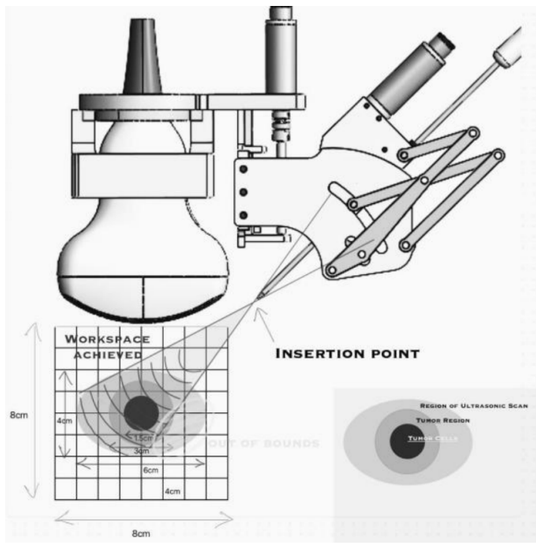


Fig. 4. Workspace achieved by the mechanism.

The workspace combines rotation about the fixed entry point with needle extension. The paper compares the swept region with the target area in the ultrasound viewing region.

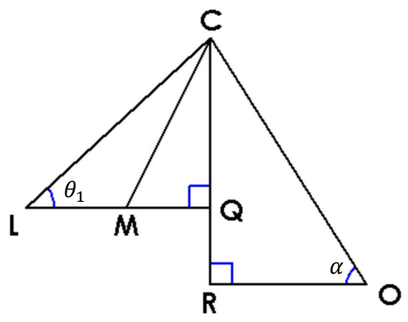
Defining of constant and variable parameters of the mechanism

The position of L can be varied with the rotation of the motor of lead screw, which will change the angle of the needle (OI) from the vertical.

Input length: - l_4, Output angle: - α

Mechanism Constraints (Does not change in operation)

AB	l_1
CK	l_2
KL	l_3
CQ	l_5
MQ	l_6
$\angle KCD$	β



Consider OR to be the vertical w.r.t the mechanism.

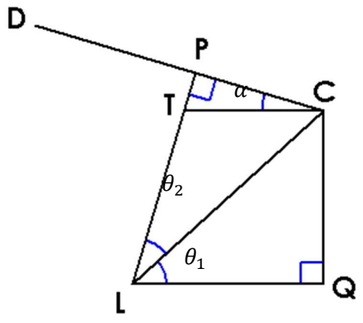
From $\Delta CRO \; and \; \Delta CQL$

$$CL\,cos\theta_1 = ML + MQ$$

$$\theta_1 = tan^{-1}\left(\frac{CQ}{ML+MQ}\right) \qquad \textit{Equation 1.}$$

$$CL = \frac{ML+MQ}{cos\,\theta_1} \qquad \textit{Equation 2.}$$

Fig 5(a). Derived $\Delta CRO \; and \; \Delta CQL$



From $\Delta PCT, \Delta TCL \; and \; \Delta CQL$

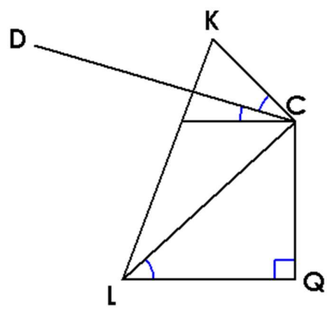
$$\angle PCL = 90^0 - \theta_2$$

$$\angle QCL = 90^0 - \theta_1$$

$$(90^0 - \theta_2) - \alpha + (90^0 - \theta_1) = 90^0$$

$$\theta_1 + \theta_2 + \alpha = 90^0 \qquad \textit{Equation 3.}$$

Fig 5(b). Derived $\Delta PCT, \Delta TCL \; and \; \Delta CQL$



From ΔKCL

Let $\angle KCL = \theta_3$

$$\theta_3 = \beta + (90^0 - \theta_2) \qquad \textit{Equation 4.}$$

$$KL^2 = KC^2 + CL^2 - 2.KC.KL.cos\theta_3$$

$$\theta_3 = cos^{-1}\left(\frac{KC^2 + CL^2 - KL^2}{2.KC.KL}\right) \qquad \textit{Equation 5.}$$

Fig 5(c). Derived ΔKCL

Calculating α from LM

1. Find $\boldsymbol{\theta_1}$ using **LM** as the input using Eqn 1.
2. Find **CL** using Eqn 2.
3. Find $\boldsymbol{\theta_3}$ using Eqn 5.
4. Find $\boldsymbol{\theta_2}$ using Eqn 4.
5. Find $\boldsymbol{\alpha}$ using Eqn 3.

Result of derivation

The inclination follows from $\theta_1 + \theta_2 + \alpha = 90°$. The equations retain the original source notation.

By Eq. 1

$$\theta_1 = \tan^{-1}\left(\frac{CQ}{ML+MQ}\right)$$

Equation 1.

By Eq. 2

$$CL = \frac{ML+MQ}{\cos\theta_1}$$

Equation 2.

By Eq. 3

$$\theta_1 + \theta_2 + \alpha = 90^\circ$$

Equation 3.

By Eq. 4

$$\theta_3 = \beta + (90^\circ - \theta_2)$$

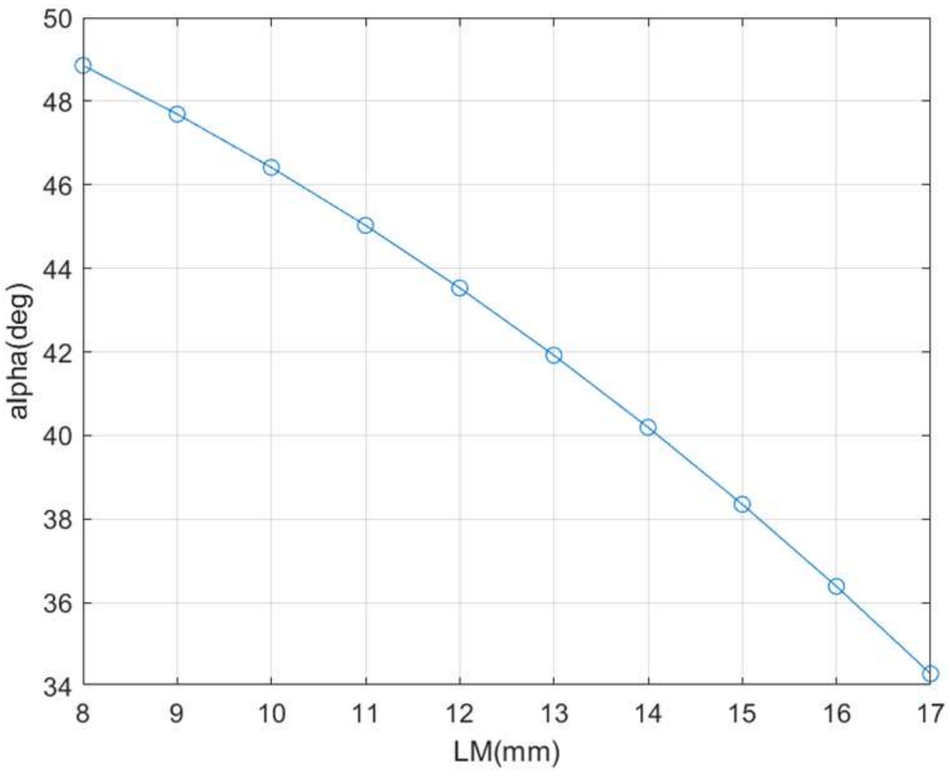
Equation 4.

By Eq. 5

$$\theta_3 = \cos^{-1}\left(\frac{KC^2 + CL^2 - KL^2}{2\,KC\,KL}\right)$$

Equation 5.

Relation Diagrams



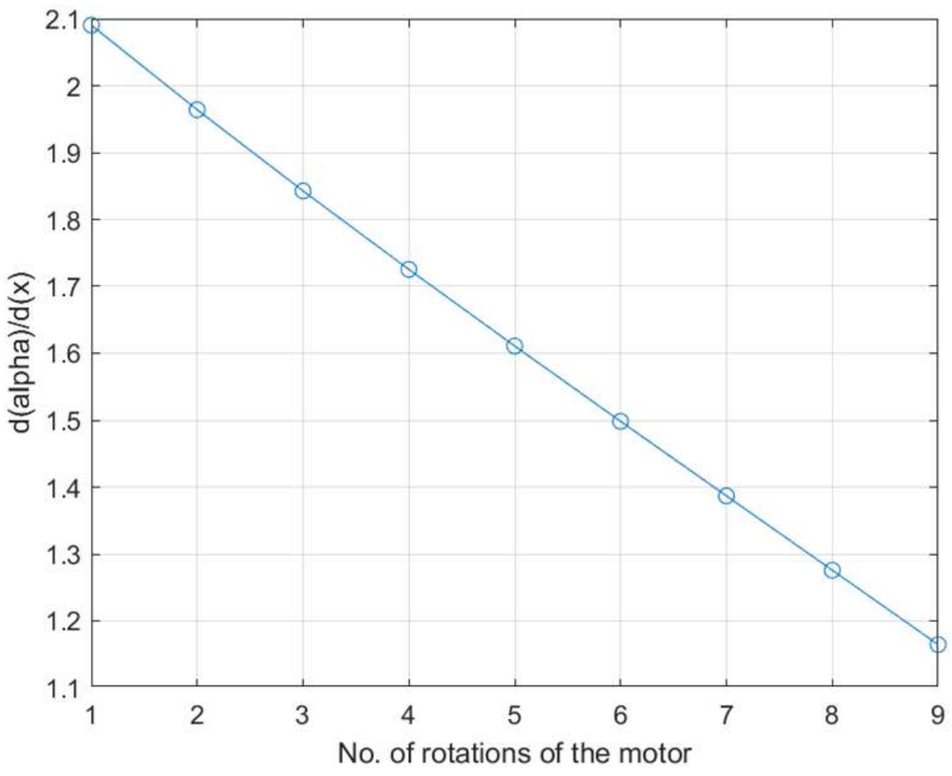


Fig 6(a). Graph: Inclination vs LM 6(b). Graph: $\frac{d\alpha}{dLM}$ vs No. of rotations of the motor (Pitch = 1mm)